

Unit 3

Network Layer

Prepared By
Ms. S. Devipriya
AP/CSE /JCE

Introduction

- The network layer in the TCP/IP protocol suite is responsible for the host-to host delivery of datagrams.
- It provides services to the transport layer and receives services from the datalink layer.
- The network layer translates the logical addresses into physical addresses.
- It determines the route from the source to the destination and also manages the traffic problems such as switching, routing and controls the congestion of data packets.
- The main role of the network layer is to move the packets from sending host to the receiving host

Services provided by network layer are

(i) PACKETIZING:

- The first duty of the network layer is definitely packetizing.
- This means encapsulating the payload (data received from upper layer) in a network-layer packet at the source and decapsulating the payload from the network-layer packet at the destination.
- The network layer is responsible for delivery of packets from a sender to a receiver without changing or using the contents.

(ii) ROUTING AND FORWARDING

Routing:

- responsible for routing the packet from its source to the destination and finding the best one among these possible routes.
- Routing is the concept of applying strategies and running routing protocols to create the decision-making tables for each router. These tables are called as routing tables.

Forwarding :

- the action applied by each router when a packet arrives at one of its interfaces.
- The decision-making table, a router normally uses for applying this action is called the forwarding table.
- When a router receives a packet from one of its attached networks, it needs to forward the packet to another attached network

(iii) Error Control, Flow control and Congestion control

Network layer does not directly provide any control mechanisms.

(iv) Security:

To provide security for a connectionless network layer, we need to have another virtual level that changes the connectionless service to a connection- oriented service. This virtual layer is called as called IPSec (IP Security).

Network Layer Performance

The performance of a network can be measured in terms of

- Delay,
- Throughput
- Packet loss

(i) Delay

A packet from its source to its destination, encounters delays. The delays in a network can be divided into four types:

- Transmission delay,
- Propagation delay,
- Processing delay and
- Queuing delay

(a) Transmission Delay:

If the first bit of the packet is put on the line at time t_1 and the last bit is put on the line at time t_2 , transmission delay of the packet is $(t_2 - t_1)$.

$$\text{Delay}_{tr} = (\text{Packet length}) / (\text{Transmission rate})$$

(b) Propagation Delay

Propagation delay is the time it takes for a bit to travel from point A to point B in the transmission media

$$\text{Delay}_{pg} = (\text{Distance}) / (\text{Propagation speed})$$

(c) Processing delay:

- The processing delay is the time required for a router or a destination host to receive a packet from its input port, remove the header, perform an error detection procedure, and deliver the packet to the output port

$$\text{Delay}_{pr} = \text{Time required to process a packet in a router or a destination host}$$

(d) Queuing Delay

The queuing delay for a packet in a router is measured as the time a packet waits in the input queue and output queue of a router.

Delay qu = The time a packet waits in input and output queues in a router

Total Delay

$$\text{Total delay} = (n + 1) (\text{Delaytr} + \text{Delaypg} + \text{Delaypr}) + (n) (\text{Delayqu})$$

(If we have n routers, we have (n +1) links.)

(ii) Throughput

- Throughput at any point in a network is defined as the number of bits passing through the point in a second, which is actually the transmission rate of data at that point.

$$\text{Throughput} = \text{minimum}\{\text{TR1}, \text{TR2}, \dots, \text{TRn}\}$$

(iii) Packet Loss

- Another issue that severely affects the performance of communication is the number of packets lost during transmission. • When a router receives a packet while processing another packet, the received packet needs to be stored in the input buffer waiting for its turn.
- A router has an input buffer with a limited size.
- A time may come when the buffer is full and the next packet needs to be dropped.
- The effect of packet loss on the Internet network layer is that the packet needs to be resent, which in turn may create overflow and cause more packet loss

IPV4 Addressing

IP address (Internet Protocol address) is a unique identifier assigned to each device connected to a network that uses the Internet Protocol for communication. It serves two main purposes:

- Identification: It uniquely identifies a device on a network.
- Location Addressing: It indicates where a device is located within a network, making data routing possible.

Internet Protocol version 4 (IPv4) is the fourth version in the development of the Internet Protocol.

Address Format

- An IPv4 address consists of 32 bit (binary digit), grouped into four section of known as **octets or bytes**. **Each octet** has 8 bits and this bits can be represented only in 0 or 1 form, and when they grouped together, they form a binary number.

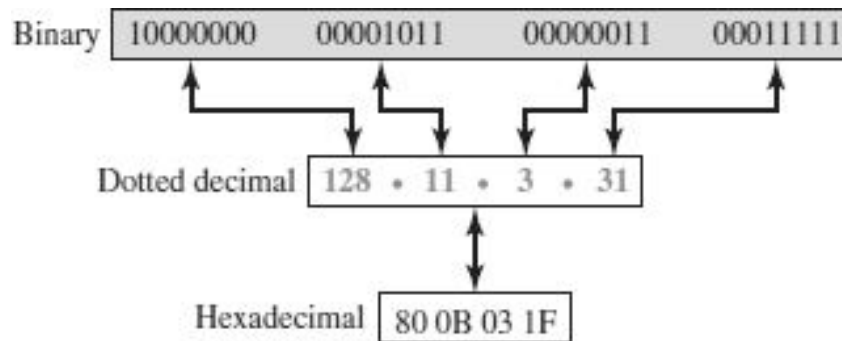
Address Space:

- If a protocol uses b bits to define an address, the address space is 2^b because each bit can have two different values (0 or 1). IPv4 uses 32-bit addresses, which means that the address space is 2^{32} or 4,294,967,296 (more than four billion).
- 4 billion devices could be connected to the Internet.

IPv4 ADDRESS NOTATION

There are three common notations to show an IPv4 address:

- (i) binary notation (base 2),
- (ii) dotted-decimal notation (base 256), and
- (iii) hexadecimal notation (base 16).



- In *binary notation*, an IPv4 address is displayed as 32 bits. To make the address more readable, one or more spaces are usually inserted between bytes (8 bits).
- In *dotted-decimal notation*, IPv4 addresses are usually written in decimal form with a decimal point (dot) separating the bytes. Each number in the dotted-decimal notation is between 0 and 255.
- In hexadecimal notation, each hexadecimal digit is equivalent to four bits. This means that a 32-bit address has 8 hexadecimal digits. This notation is often used in network programming

Parts of IPv4

IPv4 addresses consist of three parts:

- Network Part:** The network part indicates the distinctive variety that's appointed to the network. The network part conjointly identifies the category of the network that's assigned.
- Host Part:** The host part uniquely identifies the machine on your network. This part of the IPv4 address is assigned to every host. For each host on the network, the network part is the same, however, the host half must vary.
- Subnet Number:** This is the non obligatory part of IPv4. Local networks that have massive numbers of hosts are divided into subnets and subnet numbers are appointed to that.

CATEGORIES OF IPV4 ADDRESSING

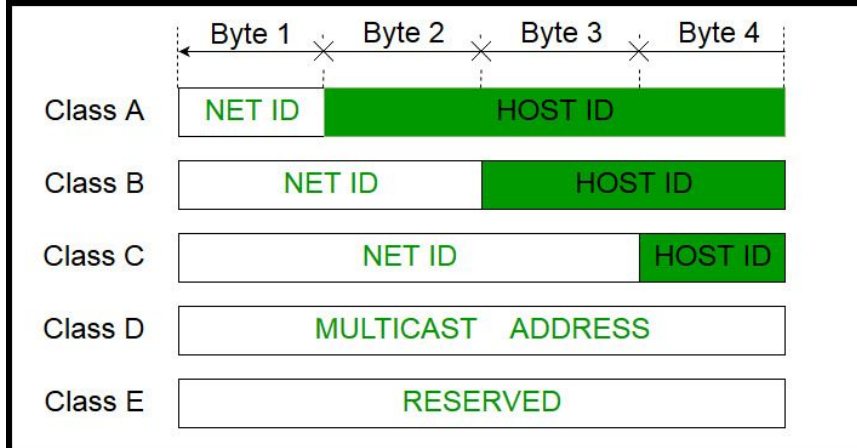
There are two broad categories of IPv4 Addressing techniques.

They are

- Classful Addressing
- Classless Addressing

Classful Addressing:

IPv4 address is divided into subclasses.



Class	Prefixes	First byte
A	$n = 8$ bits	0 to 127
B	$n = 16$ bits	128 to 191
C	$n = 24$ bits	192 to 223
D	Not applicable	224 to 239
E	Not applicable	240 to 255

Classful Addressing

Class	Higher bits	NET ID bits	HOST ID bits	No. of Networks	No. of hosts per network	Range
A	0	8	24	2^7	2^{24}	0.0.0.0 to 127.255.255.255
B	10	16	16	2^{14}	2^{16}	128.0.0.0 to 191.255.255.255
C	110	24	8	2^{21}	2^8	192.0.0.0 to 223.255.255.255
D	1110	Not Defined	Not Defined	Not Defined	Not Defined	224.0.0.0 to 239.255.255.255
E	1111	Not Defined	Not Defined	Not Defined	Not Defined	240.0.0.0 to 255.255.255.255

Class A:

In Class A, an IP address is assigned to those networks that contain a large number of hosts.

- The network ID is 8 bits long.
- The host ID is 24 bits long.
- The first bit in higher order bits of the first octet is always set to 0 and the remaining 7 bits determine the network ID.
- The 24 bits determine the host ID in any network.
- The total number of networks in Class A = $2^7 = 128$ network address
- The total number of hosts in Class A = $2^{24} - 2 = 16,777,214$ host address.



Class B:

- In Class B, an IP address is assigned to those networks that range from small- sized to large-sized networks.
- The Network ID is 16 bits long.
- The Host ID is 16 bits long.
- In Class B, the higher order bits of the first octet is always set to 10, and the remaining 14 bits determine the network ID.
- The other 16 bits determine the Host ID.
- The total number of networks in Class B = $2^{14} = 16384$ network address
- The total number of hosts in Class B = $2^{16} - 2 = 65534$ host address.



Class C:

- In Class C, an IP address is assigned to only small-sized networks.
- The Network ID is 24 bits long.
- The host ID is 8 bits long.
- In Class C, the higher order bits of the first octet is always set to 110, and the remaining 21 bits determine the network ID.
- The 8 bits of the host ID determine the host in a network.
- The total number of networks = $2^{21} = 2097152$ network address
- The total number of hosts = $2^8 - 2 = 254$ host address



Class D:

- In Class D, an IP address is reserved for multicast addresses.
- It does not possess subnetting.
- The higher order bits of the first octet is always set to 1110, and the remaining bits determines the host ID in any network.



Class E:

- In Class E, an IP address is used for the future use or for the research and development purposes.
- It does not possess any subnetting.
- The higher order bits of the first octet is always set to 1111, and the remaining bits determines the host ID in any network.



Advantage of Classful Addressing

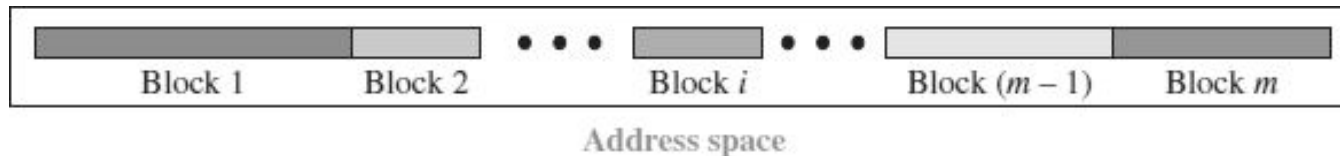
- Although classful addressing had several problems and became obsolete, it had one advantage.
- Given an address, we can easily find the class of the address and, since the prefix length for each class is fixed, we can find the prefix length immediately.
- In other words, the prefix length in classful addressing is inherent in the address; no extra information is needed to extract the prefix and the suffix.

Subnetting

- In subnetting, a class A or class B block is divided into several subnets.
- Each subnet has a larger prefix length than the original network.
- For example, if a network in class A is divided into four subnets, each subnet has a prefix of $n_{\text{sub}} = 10$.
- At the same time, if all of the addresses in a network are not used, subnetting allows the addresses to be divided among several organizations.

CLASSLESS ADDRESSING

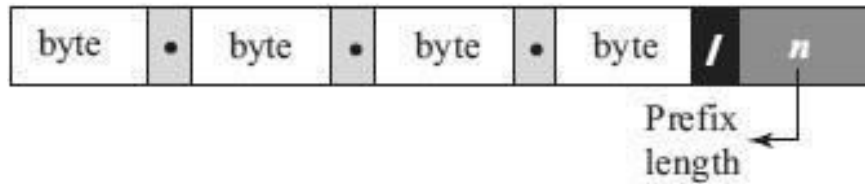
- In 1996, the Internet authorities announced a new architecture called **classless addressing**, variable-length blocks are used that belong to no classes.
- whole address space is divided into variable length blocks.
- The prefix in an address defines the block (network); the suffix defines the node (device).
- Theoretically, we can have a block of 2^0 , 2^1 , 2^2 , 2^{32} addresses.
- The number of addresses in a block needs to be a power of 2. An organization can be granted one block of addresses.



- The prefix length in classless addressing is variable that ranges from 0 to 32.
- The size of the network is inversely proportional to the length of the prefix.
- A small prefix means a larger network; a large prefix means a smaller network.
- An address in class A can be thought of as a classless address in which the
- prefix length is 8 and address in class B can be thought of as a classless address in which the prefix is 16, and so on. In other words, classful addressing is a special case of classless addressing

Notation used in Classless Addressing

- The notation used in classless addressing is informally referred to as *slash notation* and formally as *classless interdomain routing* or **CIDR**.



Examples:

12.24.76.8/8

23.14.67.92/12

220.8.24.255/25

- For example, 192.168.100.14 /**24** represents the IP address 192.168.100.14 and, its subnet mask 255.255.255.0, which has 24 leading 1-bits.
- One of the advantages of the CIDR strategy is address aggregation (sometimes called address summarization or route summarization).

Special Addresses in IPv4

- There are five special addresses that are used for special purposes:
 - ☐ this-host address,
 - ☐ limited-broadcast address,
 - ☐ loopback address,
 - ☐ private addresses,
 - ☐ multicast addresses.

Unicast Routing

- Routing is the process of selecting best paths in a network.
- In unicast routing, a packet is routed, hop by hop, from its source to its destination by the help of forwarding tables.
- Routing a packet from its source to its destination means routing the packet from a *source router* (the default router of the source host) to a *destination router* (the router connected to the destination network).
- The source host needs no forwarding table because it delivers its packet to the default router in its local network.
- The destination host needs no forwarding table either because it receives the packet from its default router in its local network.
- Only the intermediate routers in the networks need forwarding tables.

UNICAST ROUTING ALGORITHMS

- There are three main classes of routing protocols:
- **1) Distance Vector Routing Algorithm – Routing Information Protocol**
- **2) Link State Routing Algorithm – Open Shortest Path First Protocol**
- **3) Path-Vector Routing Algorithm - Border Gateway Protocol**

- Two types of Routing Protocols are used in the Internet:

1) Intradomain routing

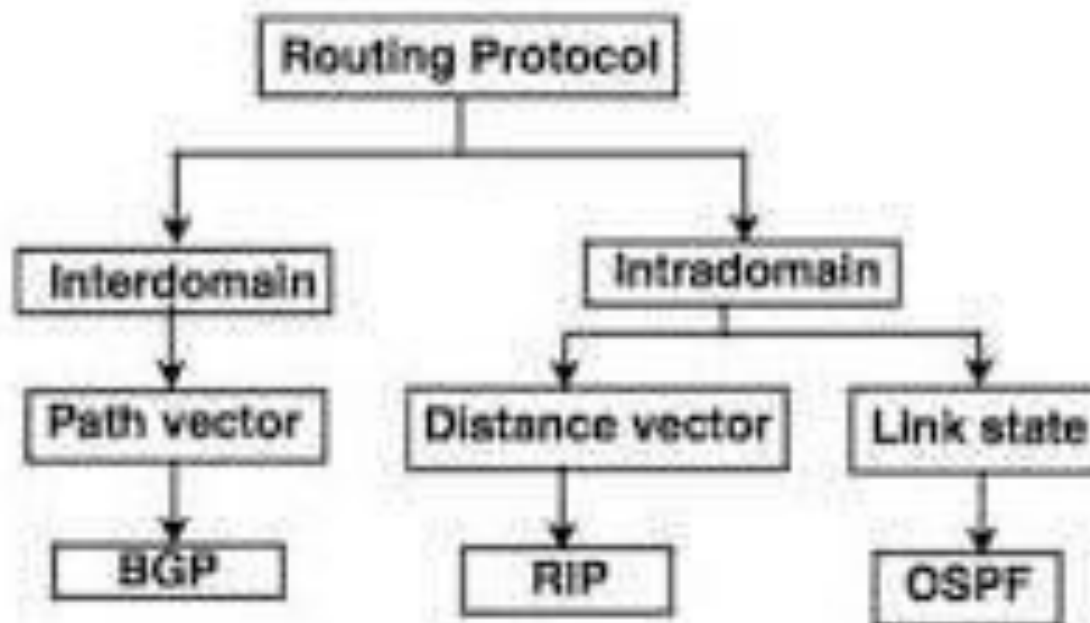
Routing within a single autonomous system

- Routing Information Protocol (RIP) - based on the distance-vector algorithm
- Open Shortest Path First (OSPF) - based on the link-state algorithm

2) Interdomain routing

Routing between autonomous systems.

- Border Gateway Protocol (BGP) - based on the path-vector algorithm



Distance Vector Routing(DSR)

- Distance Vector Routing (DVR) Protocol is a method used by routers to find the best path for data to travel across a network.
- Each router keeps a table that shows the shortest distance to every other router, based on the number of hops (or steps) needed to reach them.
- Routers share this information with their neighbors, allowing them to update their tables and find the most efficient routes.
- This protocol helps ensure that data moves quickly and smoothly through the network.

How Distance Vector Algorithm works?

- A router transmits its distance vector to each of its neighbors in a routing packet.
- Each router receives and saves the most recently received distance vector from each of its neighbors.
- A router recalculates its distance vector when:
 - It receives a distance vector from a neighbor containing different information than before.
 - It discovers that a link to a neighbor has gone down.
- From time-to-time, each node sends its own distance vector estimate to neighbors.

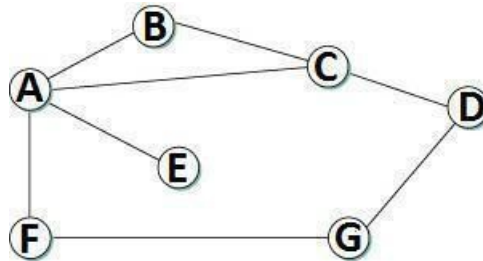
- When a node x receives new DV estimate from any neighbor v , it saves v 's distance vector and it updates its own DV using B-F equation:

$$D_x(y) = \min \{ C(x,v) + D_v(y), D_x(y) \} \text{ for each node } y \in N$$

Information kept by DV router:

- Each router has an ID
- Associated with each link connected to a router, there is a link cost (static or dynamic).
- Intermediate hops

- In given network, *cost* of each link is 1 hop. Each node sets a distance of 1 (hop) to its *immediate* neighbor and cost to itself as 0
- Distance for non-neighbors is marked as *unreachable* with value ∞ (infinity). For node *A*, nodes *B*, *C*, *E* and *F* are *reachable*, whereas nodes *D* and *G* are *unreachable*.
- There are two different circumstances under which a given node decides to send a routing update to its neighbors. (i) Periodic Update (ii) Triggered update



Node A's final routing table

Destination	Cost	NextHop
A	0	A
B	1	B
C	1	C
D	∞	—
E	1	E
F	1	F
G	∞	—

Node A's initial table

Destination	Cost	NextHop
A	1	A
B	1	B
C	0	C
D	1	D
E	∞	—
F	∞	—
G	∞	—

Node C's initial table

Destination	Cost	NextHop
A	1	A
B	∞	—
C	∞	—
D	∞	—
E	∞	—
F	0	F
G	1	G

Node F's initial table

Destination	Cost	NextHop
A	0	A
B	1	B
C	1	C
D	2	C
E	1	E
F	1	F
G	2	F

Applications of Distance Vector Routing Algorithm

- **Computer Networking** : It helps route data packets in networks.
- **Telephone Systems** : It's used in some telephone switching systems.
- **Military Applications** : It has been used to route missiles.

Advantages of Distance Vector routing

- **Shortest Path** : Distance Vector Routing finds the shortest path for data to travel in a network.
- **Usage** : It is used in local, metropolitan, and wide-area networks.
- **Easy Implementation** : The method is simple to set up and doesn't require many resources.

RIP

- Routing Information Protocol uses Distance Vector Routing to put the packets to its destination. In RIP, Each router maintains a routing table where the distance to each destination is mentioned.
- RIP uses hop count as a routing metric to find the best path between the source and the destination network.
- Routing Information Protocol is simple to implement, but it is more efficient for smaller networks, for larger networks, protocols like OSPF or EIGRP are preferred.
- RIP shares its routing tables to neighbouring routers at an interval of 30 seconds through broadcasting. If an router receives a route and it is shorter than the previous one, then router simply updates the data in the table.

- RIP has a limit of 15 hops, that is, if some route requires more than 15 hops, then that path is unreachable. In case, if a route is not updated in six successful cycles (180 seconds) in the routing table, the RIP will drop that route and inform rest of the network about the same
- Routers *advertise* the cost of reaching networks. Cost of reaching each link is 1 hop. For example, router *C* advertises to *A* that it can reach network 2, 3 at cost 0 (directly connected), networks 5, 6 at cost 1 and network 4 at cost 2.
- Each router *updates* cost and next hop for each network number.
- Infinity is defined as 16, i.e., any route cannot have more than 15 hops. Therefore RIP can be implemented on small-sized networks only

- **Command** - It indicates the packet type. Value 1 represents a request packet. Value 2 represents a response packet.
- **Version** - It indicates the RIP version number. For RIPv1, the value is 0x01.
- **Address Family Identifier** - When the value is 2, it represents the IP protocol.
- **IP Address** - It indicates the destination IP address of the route. It can be the addresses of only the natural network segment.
- **Metric** - It indicates the hop count of a route to its destination.

0	8	16	31
Command	Version	Must be zero	
Family of net 1		Address of net 1	
Address of net 1			
Distance to net 1			
Family of net 2		Address of net 2	
Address of net 2			
Distance to net 2			

Count-To-Infinity (or) Loop Instability Problem

- Suppose link from node A to E goes *down*.
- Node A advertises a distance of ∞ to E to its neighbors.
- Node B receives periodic update from C before A 's update reaches B , Node B updated by C , concludes that E can be reached in 3 hops via C .
- Node B advertises to A as 3 hops to reach E , Node A in turn updates C with a distance of 4 hops to E and so on.
- Nodes update each other until cost to E reaches *infinity*, i.e., *no convergence*.
- Routing table does not stabilize. This problem is called *loop instability* or *count to infinity*.
- When a node updates its neighbors, it does not send those routes it learned from each neighbor back to that neighbor. This *is known as split horizon*.

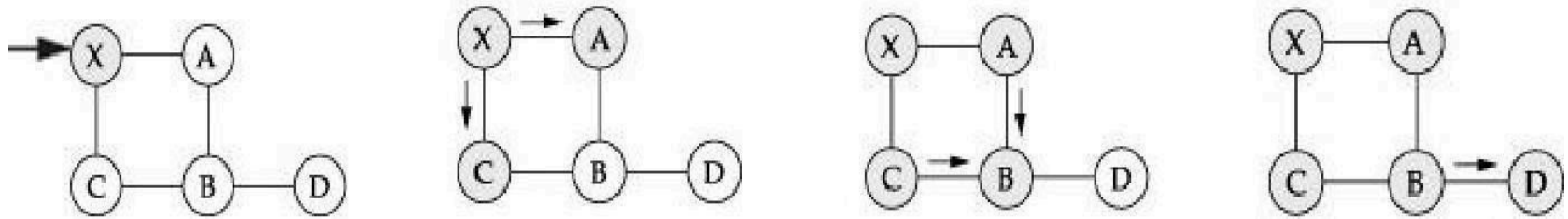
Link state Routing

- Each node knows *state* of link to its neighbors and *cost*.
- Nodes create an update packet called *link-state packet* (LSP) that contains:
 - ID of the node
 - List of neighbors for that node and associated cost
 - 64-bit Sequence number
 - Time to live
- Link-State routing protocols rely on two mechanisms:
 - Reliable flooding* of link-state information to all other nodes
 - Route calculation* from the accumulated link-state knowledge

Reliable Flooding

- Each node *sends* its LSP out on each of its directly connected links.
- When a node receives LSP of another node, checks if it has an LSP already for that node.
- If not, it stores and forwards the LSP on all other links except the incoming one.
- Else if the received LSP has a *bigger* sequence number, then it is stored and forwarded. Older LSP for that node is *discarded*.
- Otherwise discard the received LSP, since it is not latest for that node.
- Thus recent LSP of a node eventually *reaches* all nodes, i.e., reliable *flooding*

Flooding of LSP in a small network is as follows:



- When node *X* receives *Y*'s LSP (*fig a*), it floods onto its neighbors *A* and *C* (*fig b*). Nodes *A* and *C* forward it to *B*, but does not sends it back to *X* (*fig c*).
- Node *B* receives two copies of LSP with same sequence number. Accepts one LSP and forwards it to *D* (*fig d*). Flooding is complete.
- LSP is generated either *periodically* or when there is a *change* in the topology.
- Each node knows the entire topology, once it has LSP from every other node. Forward search algorithm is used to compute routing table from the received LSPs.

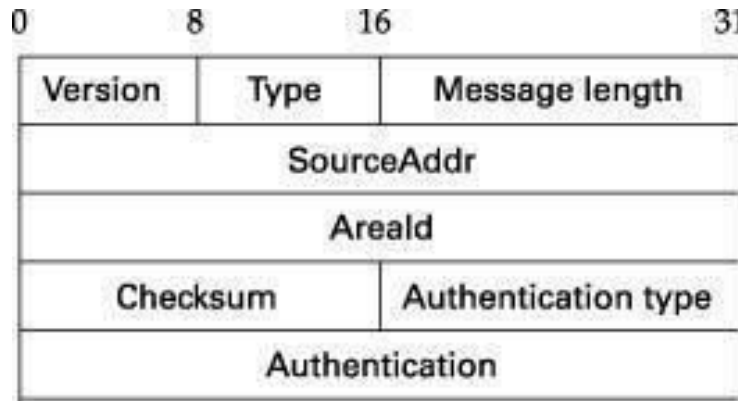
OPEN SHORTEST PATH FIRST PROTOCOL (OSPF)

- It is a widely used link-state routing protocol.

OSPF Features are:

- **Authentication**—Malicious host can collapse a network by advertising to reach every host with cost 0. Such disasters are averted by authenticating routing updates.
- **Additional hierarchy**—Domain is partitioned into areas, i.e., OSPF is more scalable.
- **Load balancing**—Multiple routes to the same place are assigned same cost. Thus traffic is distributed evenly

Link State Packet (LSP) Format



- **Version** — represents the current version, i.e., 2.
- **Type** — represents the type (1–5) of OSPF message.
- *Type 1* - “hello” message,
- *Type 2* - request,
- *Type 3* – send
- *Type 4* - acknowledge the receipt of link state messages ,
- *Type 5* - reserved
- **SourceAddr** — identifies the sender
- **Area Id** — 32-bit identifier of the area in which the node is located
- **Checksum** — 16-bit internet checksum
- **Authentication type** — 1 (simple password), 2 (cryptographic authentication).
- **Authentication** — contains password or cryptographic checksum

Difference DSR and LSR

Distance vector Routing

Each node talks only to its directly connected neighbors, but it tells them everything it has learned (i.e., distance to all nodes).

Link state Routing

Each node talks to all other nodes, but it tells them only what it knows for sure (i.e., only the state of its directly connected links).

Difference between Distance vector and Link state

Distance Vector Routing	Link State Routing
Bandwidth required is less due to local sharing, small packets and no flooding.	Bandwidth required is more due to flooding and sending of large link state packets.
Based on local knowledge, since it updates table based on information from neighbours.	Based on global knowledge, it have knowledge about entire network.
Make use of Bellman Ford Algorithm .	Make use of Dijkstra's algorithm .
Traffic is less.	Traffic is more.
Converges slowly i.e, good news spread fast and bad news spread slowly.	Converges faster.
Count of infinity problem .	No count of infinity problem.
Persistent looping problem i.e, loop will be there forever.	No persistent loops, only transient loops.
Practical implementation is RIP and IGRP .	Practical implementation is OSPF and ISIS .

Path Vector Routing(PVR)

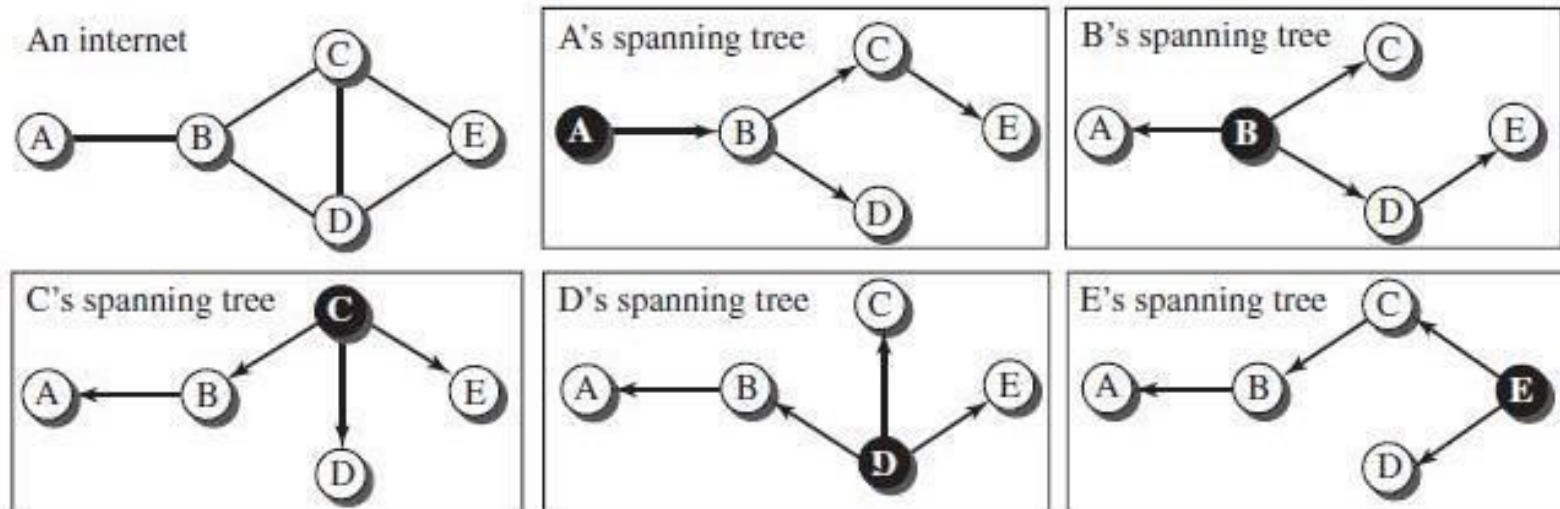
- Path-vector routing is an asynchronous and distributed routing algorithm.
- It is not based on least-cost routing, the path from a source to all destinations is determined by the *best* spanning tree which is not the least-cost tree. It is the tree determined by the source when it imposes its own policy.
- The best route is determined by the source using the policy it imposes on the route.
- One of the common policies uses the minimum number of nodes to be visited. Another common policy is to avoid some nodes as the middle node in a route.
- Path-vector routing is not actually used in an internet, and is mostly designed to route a packet between ISPs.

- The spanning trees are made, gradually and asynchronously, by each node.
- When a node is booted, it creates a *path vector* based on the information it can obtain about its immediate neighbor.
- A node sends greeting messages to its immediate neighbors to collect these pieces of information.
- Each node, after the creation of the initial path vector, sends it to all its immediate neighbors.
- Each node, when it receives a path vector from a neighbor, updates its path vector using the formula.

$$\text{Path}(x, y) = \text{best} \{ \text{Path}(x, y), [(x + \text{Path}(v, y))] \} \quad \text{for all } v\text{'s in the internet.}$$

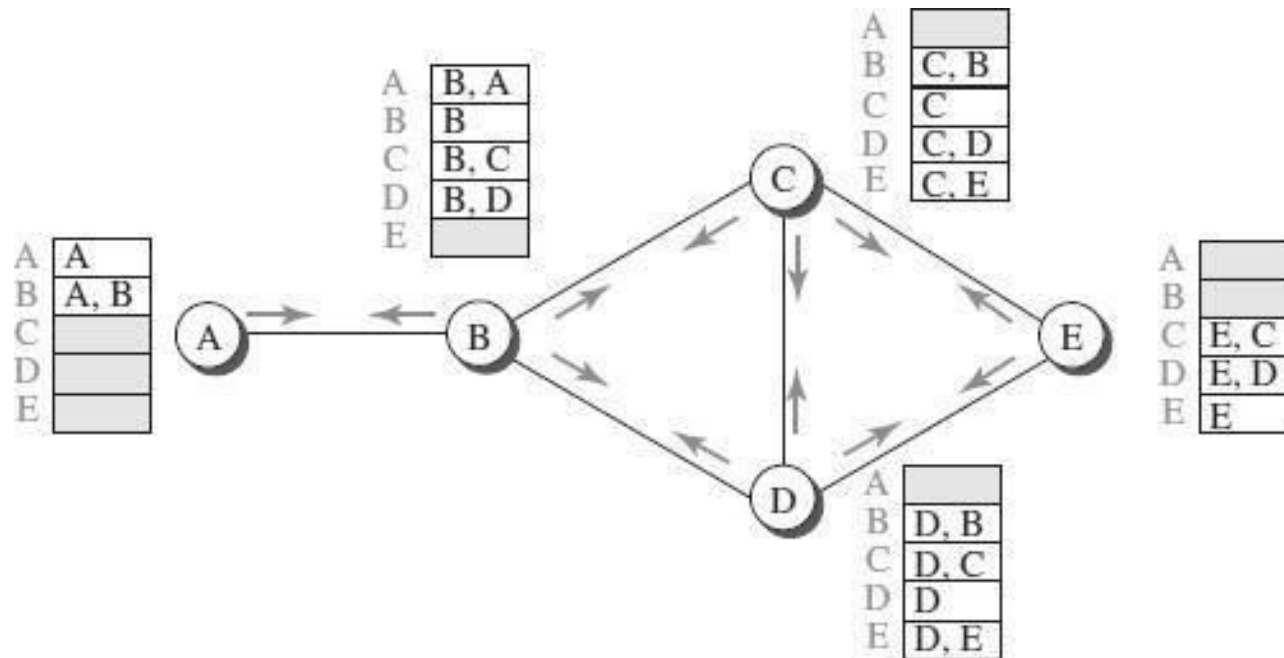
Example

- The Figure below shows a small internet with only five nodes.
- Each source has created its own spanning tree that meets its policy.
- The policy imposed by all sources is to use the minimum number of nodes to reach a destination.
- The spanning tree selected by A and E is such that the communication does not pass through D as a middle node.
- Similarly, the spanning tree selected by B is such that the communication does not pass through C as a middle node.



Path Vectors made at booting time

- Not all of these tables are created simultaneously.
- They are created when each node is booted. It shows how these path vectors are sent to immediate neighbors after they have been created.



Updating Path Vectors

- In the first event, node C receives a copy of B's vector, which improves its vector: now it knows how to reach node A.
- In the second event, node C receives a copy of D's vector, which does not change its vector.
- The vector for node C after the first event is stabilized and serves as its forwarding table.

New C		Old C		B	
A	C, B, A	A		A	B, A
B	C, B	B	C, B	B	B
C	C	C	C	C	B, C
D	C, D	D	C, D	D	B, D
E	C, E	E	C, E	E	

$C[] = \text{best}(C[], C + B[])$

Event 1: C receives a copy of B's vector

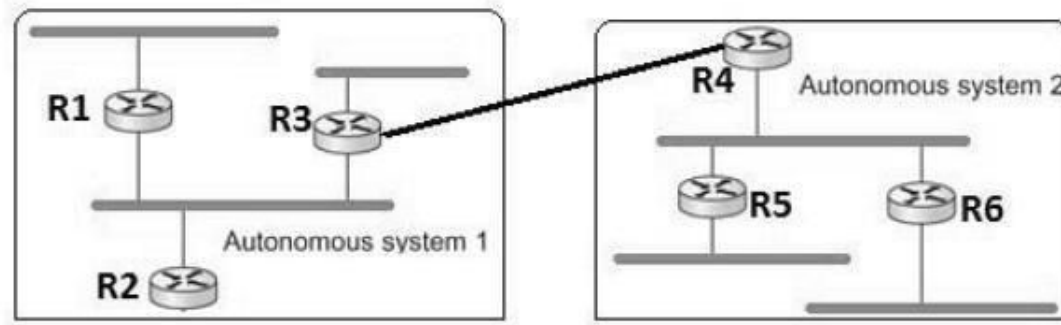
New C		Old C		D	
A	C, B, A	A	C, B, A	A	
B	C, B	B	C, B	B	D, B
C	C	C	C	C	D, C
D	C, D	D	C, D	D	D
E	C, E	E	C, E	E	D, E

$C[] = \text{best}(C[], C + D[])$

Event 2: C receives a copy of D's vector

BORDER GATEWAY PROTOCOL (BGP)

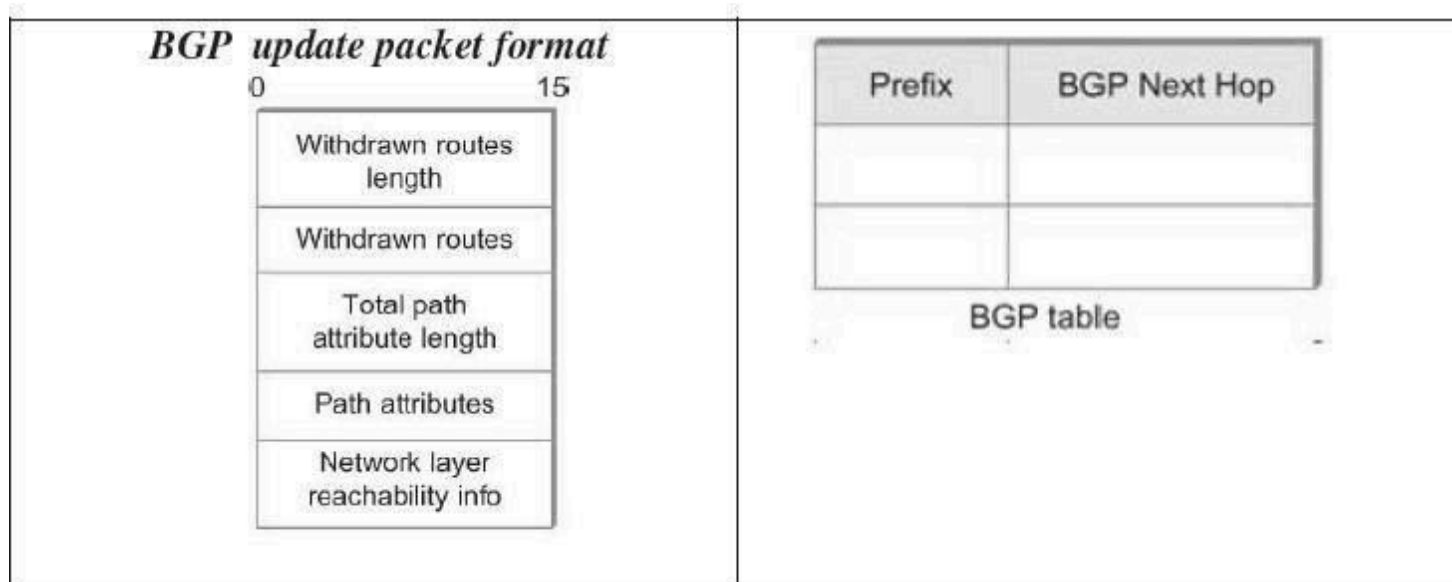
- The Border Gateway Protocol version (BGP) is the only interdomain routing protocol used in the Internet today.
- BGP4 is based on the path-vector algorithm. It provides information about the reachability of networks on the Internet.
- BGP views the internet as a set of autonomous systems interconnected arbitrarily



- Each AS has a *border router* (gateway), by which packets enter and leave that AS. In the above figure, *R3* and *R4* are border routers.
- One of the routers in each autonomous system is designated as a BGP *speaker*.

TYPES OF BGP:

- External BGP: It is used to interchange routing information between the routers in different autonomous systems, it is also known as eBGP(External Border Gateway Protocol).
- Internal BGP: It is used to interchange routing information between the routers in the same autonomous system, it is also known as iBGP(Internal Border Gateway Protocol).
- BGP speakers *exchange* reachability information with other BGP speakers, known as *external* BGP sessions.
- BGP advertises complete *path* as an enumerated list of AS (path vector) to reach a particular network. Paths must be without any *loop*, i.e., AS list is unique.



- If there are *multiple* routes to a destination, BGP speaker chooses one based on policy.
- Speakers *need not* advertise any route to a destination, even if one exists.
- Advertised paths can be *cancelled*, if a link/node on the path goes down.
- This negative advertisement is known as *withdrawn* route.
- Routes are not repeatedly sent. If there is no change, *keep alive* messages are sent.

MULTICAST ADDRESSING

- Each group has its own IP multicast address.
- IP addresses reserved for multicasting are Class D in IPv4 (Class D 224.0.0.1 to 239.255.255.255), 1111 1111 prefix in IPv6.

Two approaches in Multicasting:

In multicast routing, need to create routing trees to optimally route the packet from their source to their destination.

- Source based Tree
- Group shared tree

Source Based tree:

If there are m groups and n sources in the internet, a router needs to create $(m*n)$ routing trees. Source $\square$ root, members of the grp $\square$ leaves.

Router itself is somewhere in the tree.

Group Shared Tree:

- The designated router $\square$ also called as $\square$ “core router” (or) “Rendezvous point router” $\square$ act as the representative for the group.
- Any source that has the packet to send to a member of that group sends it to the core center and the core center is responsible for multicasting.

Multicasting

- In multicasting, there is one source and a group of destinations. Multicast supports efficient delivery to multiple destinations.
- The relationship is one to many or many-to-many.
- **One-to-Many (Source Specific Multicast) :** In this type of communication, the source address is a unicast address, but the destination address is a group address.
- Radio station broadcast
- Transmitting news, stock-price
- Software updates to multiple hosts

Many-to-Many (Any Source Multicast)

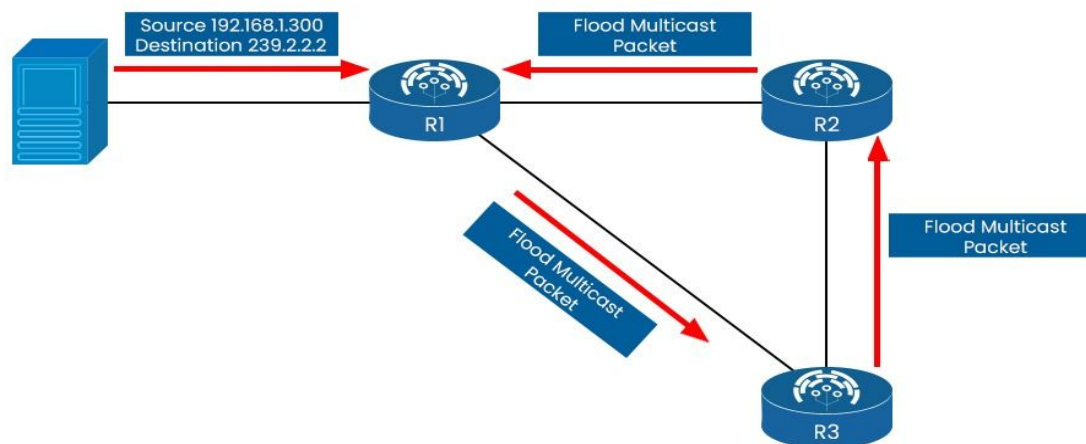
- Multimedia teleconferencing
- Online multi-player games
- Distributed simulations

- Multicast routing protocols are classified into two categories:
(a) dense mode and (b) sparse mode.

Dense-mode protocols assume that most nodes in the network are interested in receiving multicast data, while sparse-mode protocols assume that only a few nodes are interested.

Dense Mode

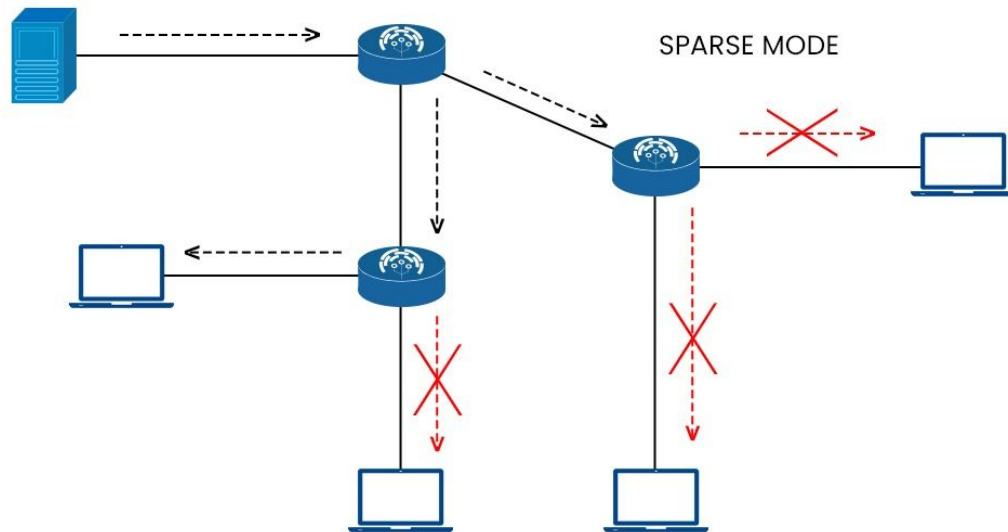
Dense mode protocols assume that there are many receivers in the network and use flooding to send data packets to all nodes. Then, nodes that are not interested in receiving the data prune themselves from the multicast tree by sending messages to their upstream routers. This way, only the nodes that want to receive the data remain in the tree. Some dense mode protocols are DVMRP, MOSPF, PIM-DM.



Sparse Mode

Sparse mode protocols assume that there are few receivers in the network and use explicit join messages to build multicast trees. They use a rendezvous point (RP) as a central node that connects sources and receivers. Receivers send join messages to the RP, which then forwards them to the sources. Sources send data packets to the RP, which then forwards them to the receivers.

One of the most common sparse-mode protocols is PIM sparse mode.



Intra Domain Multicast Protocols:

Extension of unicast routing protocols using source based tree approach.

(i) DVMRP □ extension of RIP

(ii) MOSPF □ extension of OSPF

Independent protocol : can be used in two approaches Source based and Shared group. □ PIM

Multicast Distance Vector (DVMRP)

Multicast distance vector routing uses source-based trees, but the router never actually makes a routing table. When a router receives a multicast packet, it forwards the packet as though it is consulting a routing table and create source based multicast tree using following steps.

Step 1: Flooding

- Router on receiving a multicast packet from source S to a Destination from NextHop, forwards the packet on all out-going links.
- Packet is flooded and looped back to S.
- The drawbacks are:
 - o It floods a network, even if it has no members for that group.
 - o Packets are forwarded by each router connected to a LAN, i.e., duplicate flooding

Step 2: The router uses an algorithm called “**Reverse Path Forwarding (RPF)**” to simulate creating part of optimal source based tree between the source and itself.

- RPF eliminates the looping problem in the flooding process.
- Only one copy is forwarded and the other copies are discarded.
- RPF forces the router to forward a multicast packet from one specific interface:

the one which has come through the shortest path from the source to the router.

- Packet is flooded but not looped back to S

- **Step 3:** The router uses an algorithm called “**Reverse path broadcasting (RPB)**” to create a broadcast spanning tree whose root is the router itself and whose leaves are all networks in the internet.
- RPB does not multicast the packet, it broadcasts it.
- RPB creates a shortest path broadcast tree from the source to each destination.
- It guarantees that each destination receives one and only one copy of the packet.
- We need to prevent each network from receiving more than one copy of the packet.
- If a network is connected to more than one router, it may receive a copy of the packet from each router.
- One router identified as parent called designated Router (DR).
- Only parent router *forwards* multicast packets from source *S to the attached network*.
- When a router that is not the parent of the attached network receives a multicast packet, it simply drops the packet.

Step 4: The router uses an algorithm called **“Reverse path multicasting (RPM)”** to create a multicast tree by cutting some branches of the tree that end in networks with no member in the group.

- To increase efficiency, the multicast packet must reach only those networks that have active members for that particular group.
- RPM adds pruning and grafting to RPB to create a multicast shortest path tree that supports dynamic membership changes.

Pruning:

- Sent from routers receiving multicast traffic for which they have no active group members “Prunes” the tree created by DVMRP
- Stops needless data from being sent

Grafting:

- Used after a branch has been pruned back
- Sent by a router that has a host that joins a multicast group
- Goes from router to router until a router active on the multicast group is reached

Sent for the following cases

- A new host member joins a group
- A new dependent router joins a pruned branch
- A dependent router restarts on a pruned branch

Multicast Linkstate (MOSPF)

- Extended OSPF Protocol.
- Uses Source based tree approach.
- The protocol requires a new link state update packet to associate the unicast address of a host with the group address or addresses the host is sponsoring. This packet is called the group-membership LSA.

Protocol Independent Multicast (PIM)

- PIM divides multicast routing problem into *sparse* and *dense* mode.
- PIM sparse mode (PIM-SM) is widely used.
- PIM does not rely on any type of unicast routing protocol, hence protocol independent.
- Routers explicitly join and leave multicast group using ***Join and Prune messages***.
- One of the router is designated as *rendezvous point* (RP) for each group in a domain to receive PIM messages.
- Multicast forwarding *tree* is built as a result of routers sending Join messages to RP.
- Two types of trees to be constructed:
 - ***Shared tree*** - used by all senders
 - ***Source-specific tree*** - used only by a specific sending host
- The normal mode of operation creates the shared tree first, followed by one or more source-specific trees

PIM-DM

- PIM-DM is used when there is a possibility that each router is involved in multicasting (dense mode).
- In this environment, the use of a protocol that broadcasts the packet is justified because almost all routers are involved in the process.
- PIM-DM is a source-based tree routing protocol that uses RPF and pruning and grafting strategies for multicasting. Its operation is like that of DVMRP.

PIM-SM

- PIM-SM is used when there is a slight possibility that each router is involved in multicasting (sparse mode).
- PIM-SM is used in a sparse multicast environment such as a WAN.
- It is a group-shared tree routing protocol that has a rendezvous point (RP) as the source of the tree.